Lab Exercise 20 - Deploying a Web Application using AWS ECR and ECS

Viraj Bhidola
500121825
B2 DevOps

Objective

Learn how to containerize a web application, push the Docker image to Amazon Elastic Container Registry (ECR), and deploy it as a highly available service using Amazon Elastic Container Service (ECS) with AWS Fargate.

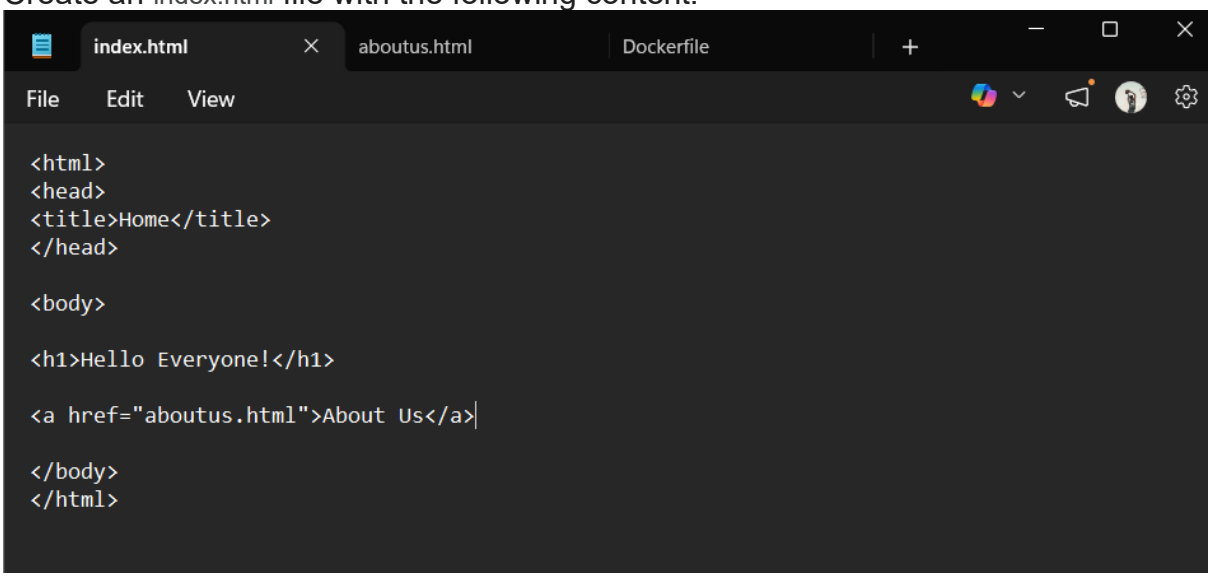
Prerequisites

- AWS Account with administrative privileges.
- AWS CLI installed and configured on your local machine.
- Docker installed and running.
- Basic understanding of Dockerfiles and HTML.

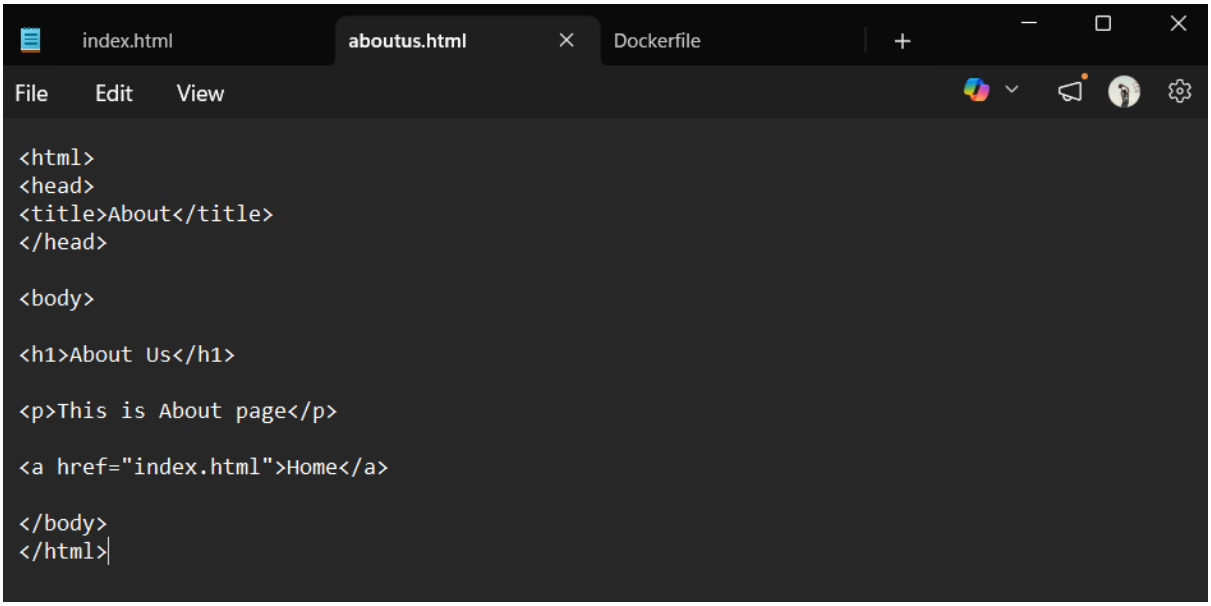
Step 1: Create the Web Application Files

First, prepare the source code and Dockerfile for your web application.

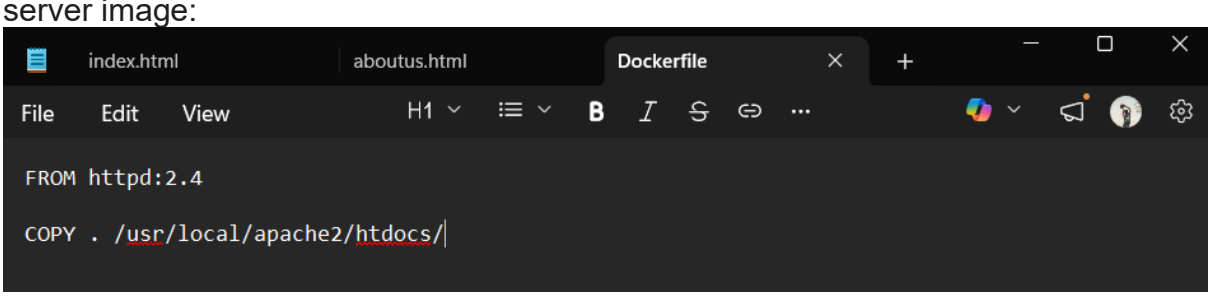
1. Create a directory for your project (e.g., my-web-app) and navigate into it.
2. Create an index.html file with the following content:



3. Create an aboutus.html file with the following content:



4. Create a Dockerfile to serve the HTML files using the official Apache HTTP server image:



Step 2: Create an ECR Repository

Create a secure place in AWS to store your Docker images.

1. Log in to the AWS Management Console and navigate to **Amazon Elastic Container Registry (ECR)**.
2. Click **Create repository**.
3. Set the repository visibility to **Private**.
4. Enter lab20 as the **Repository name**.
5. Leave the remaining settings as default (e.g., Mutable tags, AES-256 encryption) and click **Create repository**.
6. Once created, note down the **URI** of your new repository (e.g., 578673727059.dkr.ecr.ap-south-1.amazonaws.com/lab20).

Containers

Amazon Elastic Container Registry

Share and deploy container software, publicly or privately

Amazon Elastic Container Registry (ECR) is a fully managed container registry that makes it easy to store, manage, share and deploy your container images and artifacts anywhere.

Create a repository

Create

Pricing (US)

You only pay for the amount of data you store in your public or private repositories and data transferred to the Internet.

ECR pricing

Enable private connection to ECR private repositories using VPC interface endpoints and reduce costs.

Find out more
Deploy VPC interface endpoints

Getting started

What is Amazon ECR?
Getting started with ECR
Set up a CICD pipeline with ECR

How it works

Benefits

Fully managed

Secure

Amazon ECR > Private registry > Repositories > Create a private repository

General settings

Repository name

Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.

578673727059.dkr.ecr.ap-south-1.amazonaws.com/ lab20

5 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers and special characters ._-/.

Image tag settings Info

Image tag mutability

Choose the tag mutability setting.

- Mutable
  Image tags can be overwritten
- Immutable
  Image tags can't be overwritten.

Mutable tag exclusions

Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.

Add filter

Filters must only contain letters, numbers, and special characters (._*-). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.

Encryption settings Info

⚠ The encryption settings for a repository can't be changed once the repository is created.

Encryption configuration

By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.

- AES-256
  Industry standard Advanced Encryption Standard (AES) encryption
- AWS KMS
  AWS Key Management Service (KMS)

Amazon ECR > Private registry > Repositories

Amazon Elastic Container Service

- Private registry
  - Repositories
  - Features & Settings
- Public registry
  - Repositories
  - Settings

ECR public gallery
Amazon ECS
Amazon EKS

Getting started
Documentation

✓ Successfully created private repository, lab20

Private repositories (1)

View push commands | Delete | Actions | Create repository

Search by repository substring

Repository name	URI	Created at	Tag im y
lab20	071179620453.dkr.ecr.eu-north-1.amazonaws.com/lab20	April 17, 2026, 01:36:44 (UTC+05.5)	Mu

Step 3: Build and Push the Docker Image

Use your local terminal to build the image and push it to the ECR repository created in Step 2.

1. Authenticate your local Docker client to your ECR registry. Replace the URI with your specific account details:

   aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 578673727059.dkr.ecr.ap-south-1.amazonaws.com

   Expected Output: Login Succeeded

2. Build your Docker image locally:

   docker build -t lab20 .

3. Tag your local image to match your ECR repository URI:

   docker tag lab20:latest 578673727059.dkr.ecr.ap-south-1.amazonaws.com/lab20:latest

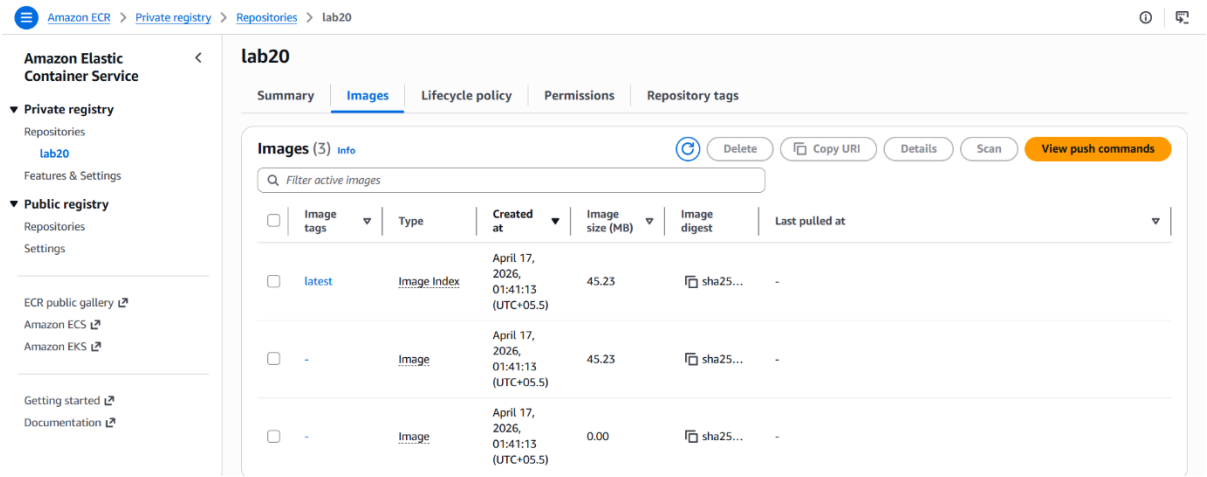
4. Push the tagged image to AWS ECR:

   docker push 578673727059.dkr.ecr.ap-south-1.amazonaws.com/lab20:latest

```
C:\Users\ASUS\my-web-app>docker build -t lab20 .
[+] Building 13.6s (8/8) FINISHED                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                        0.1s
 => => transferring dockerfile: 88B                                         0.0s
 => [internal] load metadata for docker.io/library/httpd:2.4                4.4s
 => [auth] library/httpd:pull token for registry-1.docker.io                0.0s
 => [internal] load .dockerignore                                           0.1s
 => => transferring context: 2B                                             0.0s
 => [internal] load build context                                           0.1s
 => => transferring context: 471B                                           0.0s
 => [1/2] FROM docker.io/library/httpd:2.4@sha256:bdba5c86022f2d6ad07       8.2s
 => => resolve docker.io/library/httpd:2.4@sha256:bdba5c86022f2d6ad07       0.1s
 => => sha256:a838a4be55ced5f2d7208c56a3a13ecbd151105 2.00MB / 2.00MB       1.0s
 => => sha256:4e51448030490b8604d7662b05848e89e3563fe3b95 289B / 289B       1.1s
 => => sha256:8261ede62f8c1c85c3e4b2ae3a9ae5a586b84fcc61b 146B / 146B       1.4s
 => => sha256:61e1798e23ba12f14505e225391338db290ee 13.45MB / 13.45MB       2.6s
 => => sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776 29.78MB / 29.78MB       2.8s
 => => extracting sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a60       2.6s
 => => extracting sha256:8261ede62f8c1c85c3e4b2ae3a9ae5a586b84fcc61b6       0.0s
 => => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb55774       0.0s
 => => extracting sha256:a838a4be55ced5f2d7208c56a3a13ecbd15110574ee3       0.5s
 => => extracting sha256:61e1798e23ba12f14505e225391338db290ee6cecfb6       0.8s
 => => extracting sha256:4e51448030490b8604d7662b05848e89e3563fe3b955       0.0s
 => [2/2] COPY . /usr/local/apache2/htdocs/                                 0.6s
 => exporting to image                                                      0.6s
 => => exporting layers                                                     0.2s
 => => exporting manifest sha256:7596d625662208c46cf99636d5d7f78336ba       0.0s
 => => exporting config sha256:e0bf7fb5087aab0e41a4d1da7156ff61f52f7f       0.0s
 => => exporting attestation manifest sha256:8c8b43c4acafea5d73bd7d8a       0.1s
 => => exporting manifest list sha256:bc49c487059a3770ceb1364dc2ea7fc       0.0s
 => => naming to docker.io/library/lab20:latest                             0.0s
 => => unpacking to docker.io/library/lab20:latest                          0.1s

C:\Users\ASUS\my-web-app>
```

```
C:\Users\ASUS\my-web-app>aws ecr get-login-password --region eu-north-1 | do
cker login --username AWS --password-stdin 071179620453.dkr.ecr.eu-north-1.a
mazonaws.com
Login Succeeded
```

```
C:\Users\ASUS\my-web-app>docker tag lab20:latest 071179620453.dkr.ecr.eu-nor
th-1.amazonaws.com/lab20:latest

C:\Users\ASUS\my-web-app>docker push 071179620453.dkr.ecr.eu-north-1.amazona
ws.com/lab20:latest
The push refers to repository [071179620453.dkr.ecr.eu-north-1.amazonaws.com
/lab20]
5435b2dcdf5c: Pushed
8261ede62f8c: Pushed
a838a4be55ce: Pushed
4e5144803049: Pushed
740c98f8f440: Pushed
8d13eb37956e: Pushed
4f4fb700ef54: Pushed
61e1798e23ba: Pushed
latest: digest: sha256:bc49c487059a3770ceb1364dc2ea7fc8dc383002e39078dc3282b
820163eebe5 size: 856

C:\Users\ASUS\my-web-app>
```



Step 4: Create an ECS Cluster

Set up the infrastructure to run your containers.

1. Navigate to **Amazon Elastic Container Service (ECS)** in the AWS Console.
2. Select **Clusters** from the left-hand menu and click **Create cluster**.
3. Set the **Cluster name** to lab20.
4. Under Infrastructure, select **AWS Fargate (Serverless)** to run containers without managing the underlying EC2 instances.
5. Click **Create**.

Amazon Elastic Container Service

Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions New
Account settings

AWS Batch
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Containers

Amazon Elastic Container Service
Fully managed containers

Amazon Elastic Container Service (Amazon ECS) is a highly scalable and fast container management service that makes it easy to run, stop and manage containers on a cluster.

Deploy your containerised applications

Amazon ECS makes it easy to deploy, manage and scale Docker containers running applications, services and batch processes.

Get started

How it works

Pricing (US)

Amazon ECS on AWS Fargate
Fargate pricing

Amazon ECS Managed Instances
ECS Managed Instances pricing

Amazon ECS on Amazon EC2 EC2 pricing

Amazon ECS Anywhere
ECS Anywhere pricing

More resources

Documentation

Amazon Elastic Container Service > Clusters

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Introducing managed daemons
You can now manage software agents as standalone daemons across container instances in a capacity provider, independent of your applications. Find out more

Create daemon task definition

Clusters (0) Info

Last updated 15 April 2026, 23:34 (UTC+5:30)

Create cluster

Search clusters

By default, we only load up to 1,000 clusters at a time.

Cluster	Services	Tasks	Container instances	CloudWatch monitoring	Capacity provider strategy

No clusters

Create cluster

Amazon Elastic Container Service > Create cluster

Amazon Elastic Container Service

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Tell us what you think

Create cluster Info

An Amazon ECS cluster groups together tasks and services, and allows for shared capacity and common configurations. All of your tasks, services and capacity must belong to a cluster.

Cluster configuration

Cluster name

lab20

Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

▸ **Service Connect defaults** – *optional*

▾ Infrastructure - *advanced* Info

Configure the manner of obtaining compute resources that will be used to host your application.

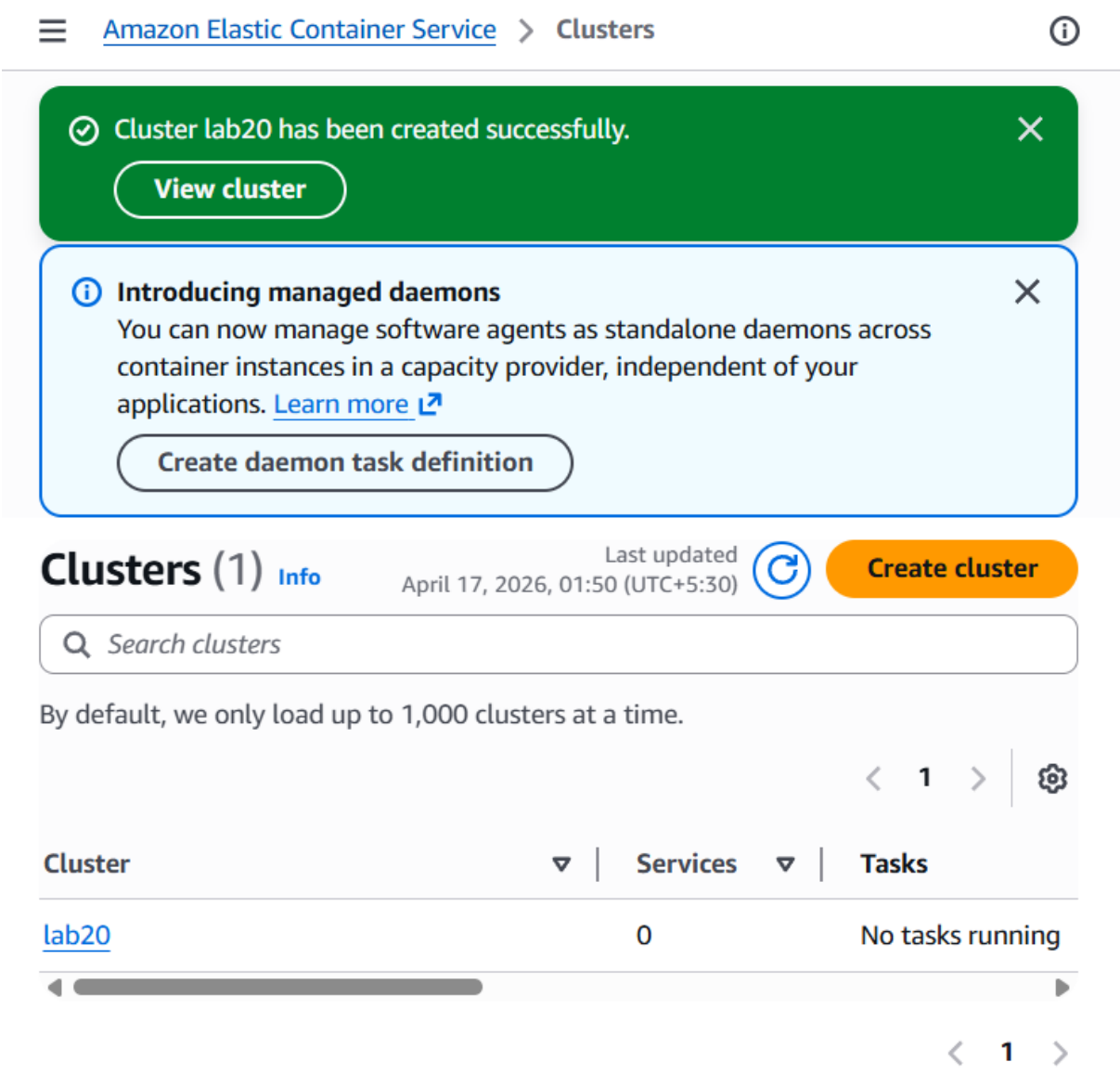
Select a method of obtaining compute capacity

Your cluster is automatically configured for AWS Fargate (serverless), but you may choose to add Amazon EC2 instances (servers).

- (●) **Fargate only** — Serverless – you don't think about creating or managing servers. Great for most common workloads.
- ( ) **Fargate and managed instances** — Managed instances - Amazon ECS will manage patching and scaling on your behalf while giving you configurability about the types of instances. Great for more advanced workloads.
- ( ) **Fargate and Self-managed instances** — Self-managed instances - you must ensure the instances are patched and scaled properly, and you have full control over the instances.

▸ **Monitoring** - *optional*
Configure observability, encryption and logging options to maintain compliance and operational visibility of your container environment.

▸ **Encryption** - *optional*
Choose the KMS keys used by tasks running in this cluster to encrypt your storage.



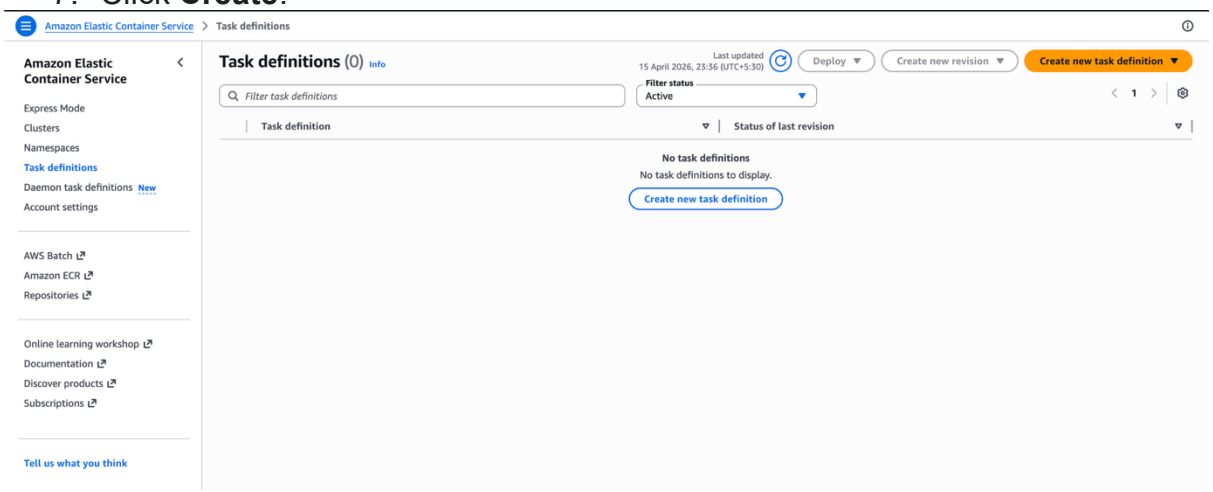
Step 5: Create a Task Definition

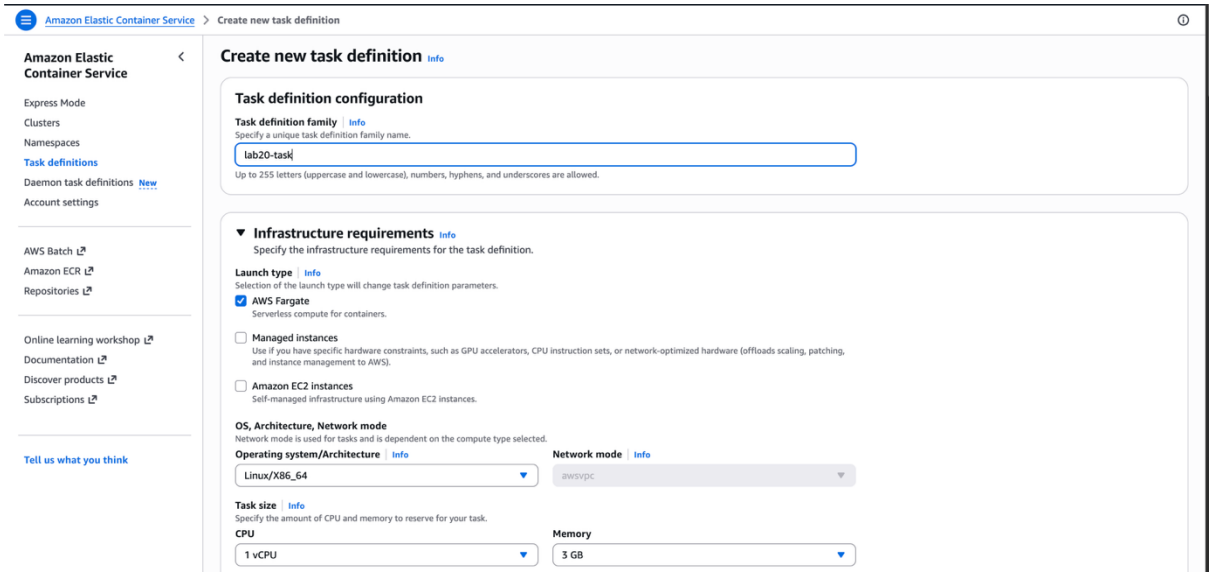
Define how your container should be configured when it runs.

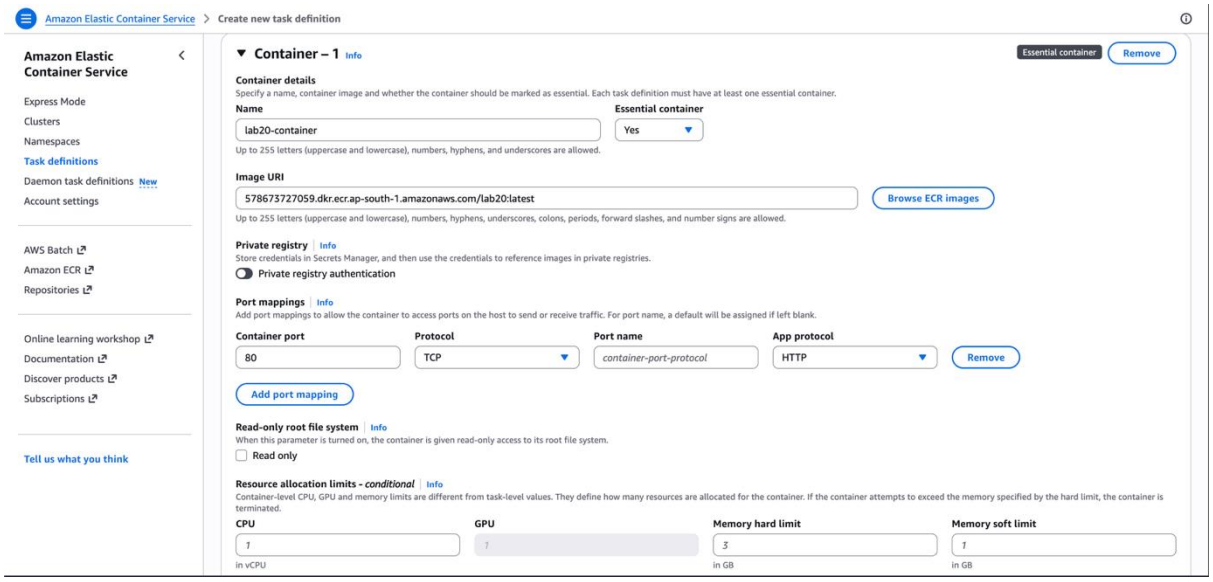
1. In ECS, select **Task definitions** from the left menu and click **Create new task definition**.
2. Set the **Task definition family** name to lab20-task.
3. Under Infrastructure requirements, select **AWS Fargate** as the launch type.
4. Set the Operating system/Architecture to **Linux/X86_64**.
5. Configure the Task size:
   - **CPU:** 1 vCPU
   - **Memory:** 3 GB
6. Under Container - 1, configure the following:
   - **Name:** lab20-container
   - **Image URI:** Paste the URI of your ECR image (e.g., 578673727059.dkr.ecr.ap-south-1.amazonaws.com/lab20:latest).

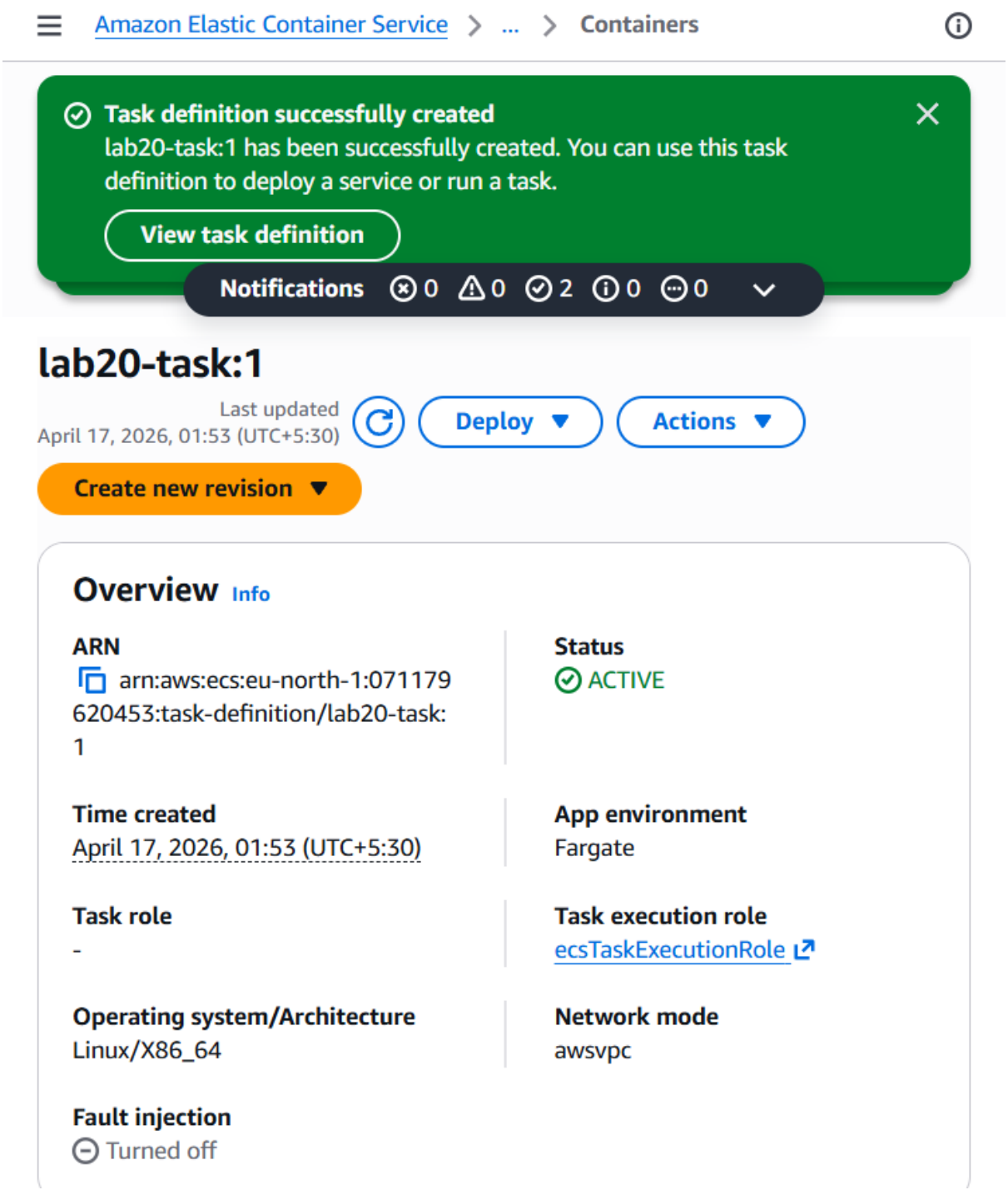
- **Container port:** 80
- **Protocol:** TCP

7. Click **Create**.







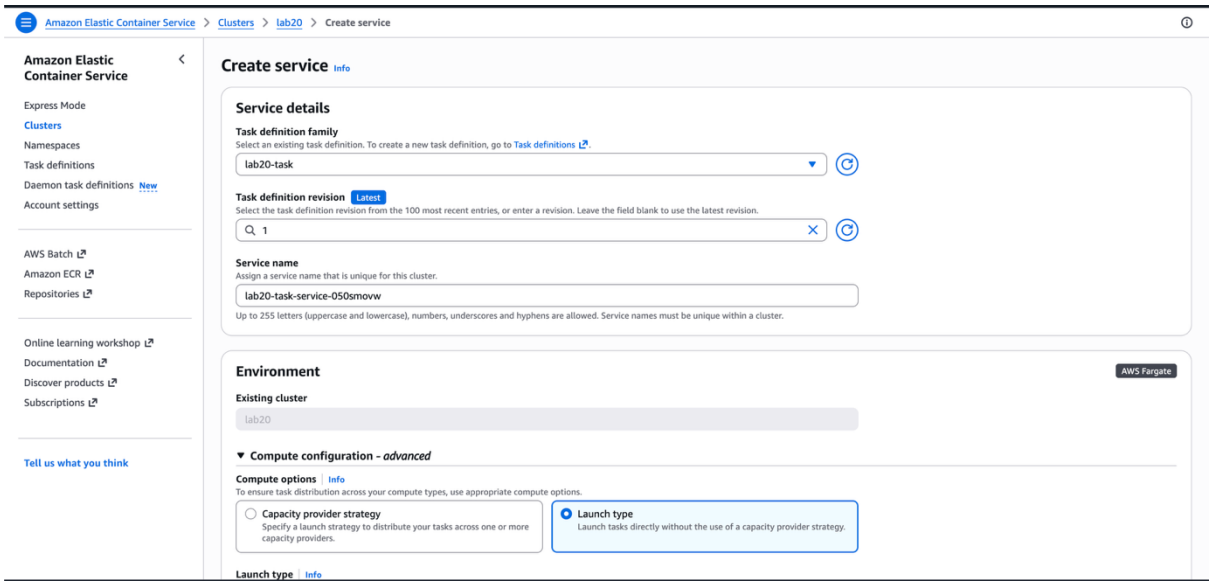


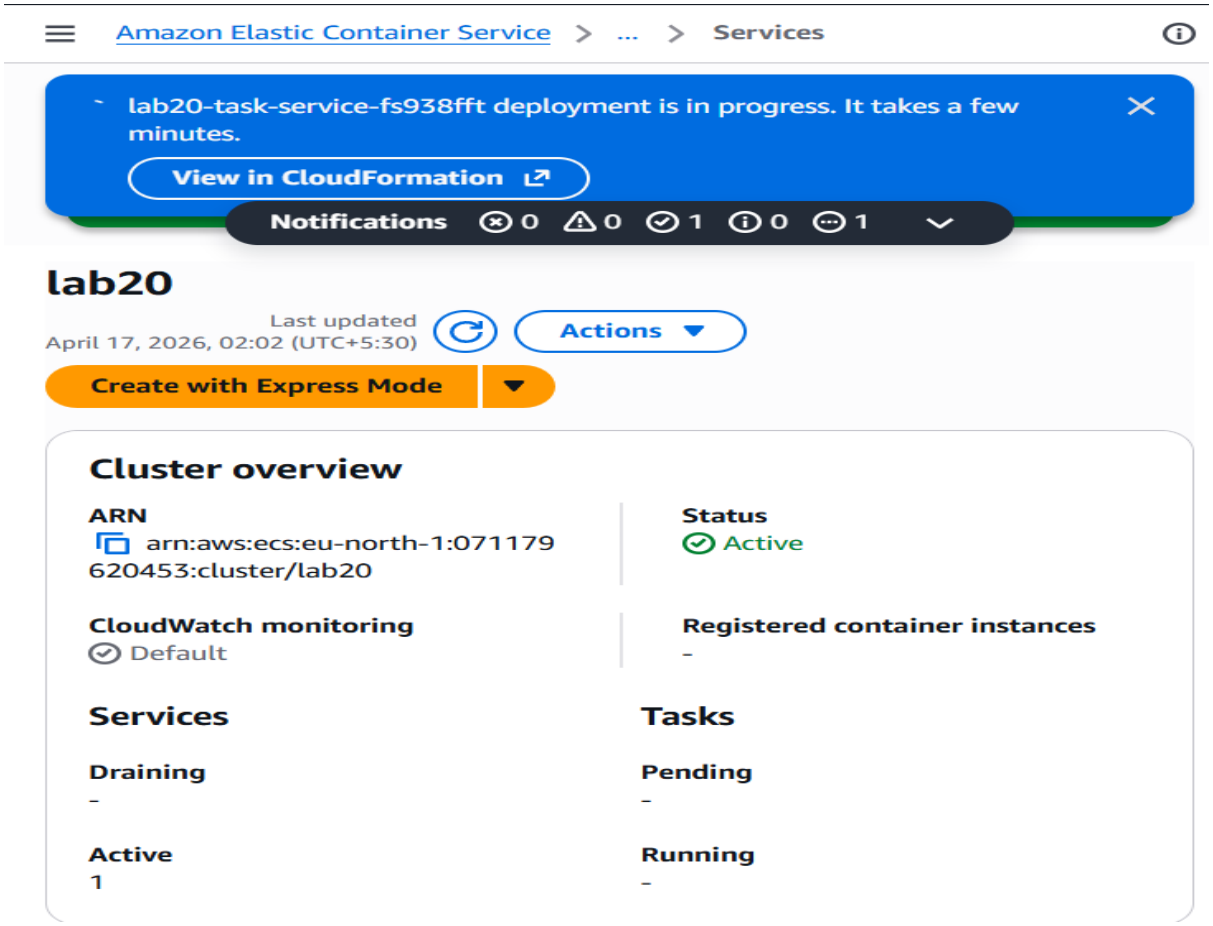
Step 6: Create an ECS Service

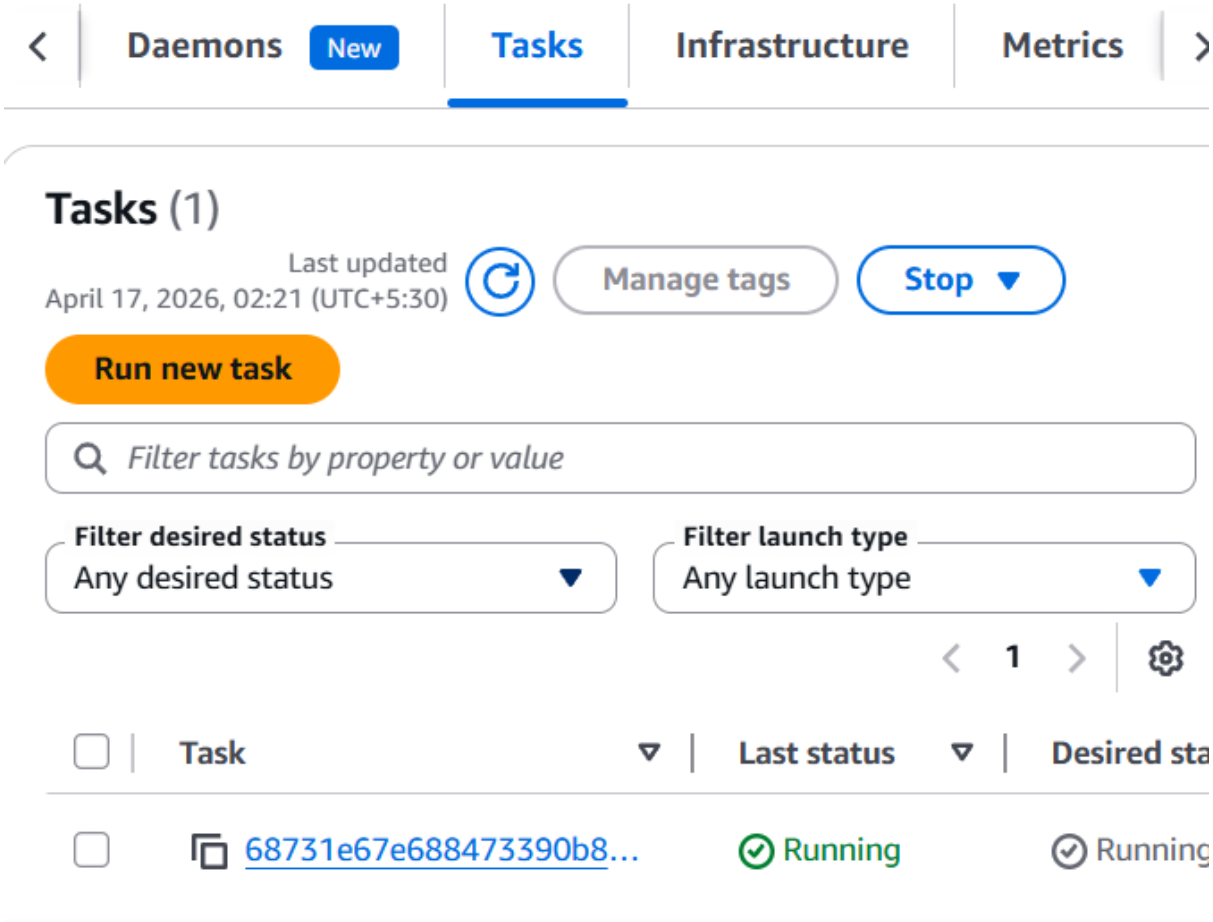
Deploy your task definition into the cluster.

1. Navigate back to your lab20 cluster and go to the **Services** tab. Click **Create**.
2. Under Compute configuration, select **Launch type** and ensure **Fargate** is chosen.

3. Under Deployment configuration, set the **Task definition family** to lab20-task and select the latest revision.
4. Assign a **Service name**, such as lab20-task-service-050smovw.
5. Set the desired tasks to 1.
6. Click **Create** to deploy the service. Wait a few minutes for the deployment to finish and the task status to turn to **Running**.



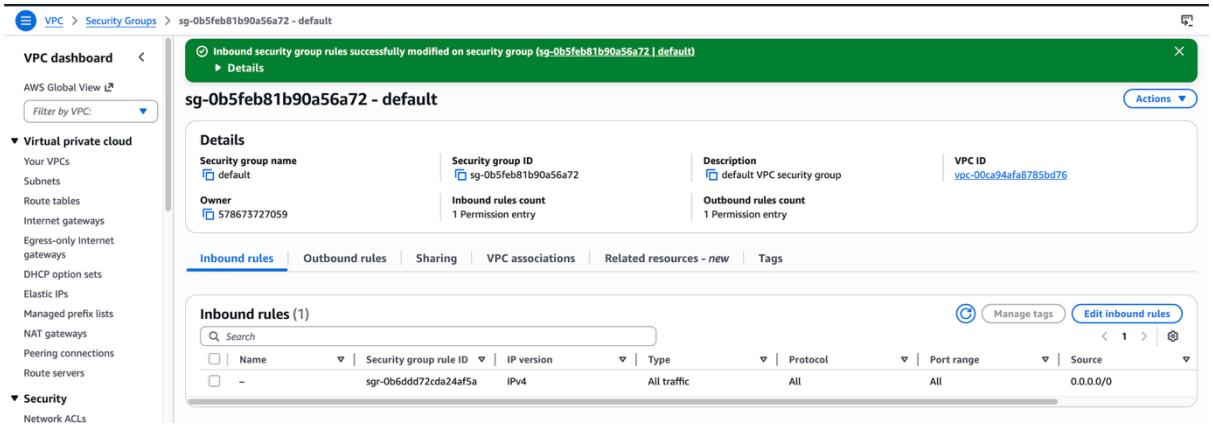




Step 7: Configure Security Groups

Ensure public traffic can reach your web server on port 80.

1. Locate the Elastic Network Interface (ENI) or the Security Group associated with your running ECS Task.
2. Open the **VPC Dashboard** or EC2 Security Groups console.
3. Select the relevant security group (e.g., sg-0b5feb81b90a56a72 - default).
4. Edit the **Inbound rules**.
5. Add a rule to allow **All traffic** (or specifically HTTP) on Port **80** from the Source **0.0.0.0/0**.
6. Save the rules.



Step 8: Verify the Deployment

Access your live application over the internet.

1. In the ECS console, navigate to your running task inside the lab20 cluster.
2. Find the **Public IP** address assigned to the Fargate task
3. Open a new tab in your web browser and navigate to (http://13.127.18.163).
4. You should see the "Hello Everyone!" heading. Click the link to verify routing to the "About Us" page.